

VBF-T1R1NOFORCE-DFMEA-01 - Design FMEA (qualitative version)

VBF-T1R1NOFORCE-DFMEA-01 · Generation date: 2026-08-25 · Qualitative ratings based on simulation signals; severity/occurrence three-level (high/medium/low); RPN = simplified SxO matrix (qualitative caliber). Complete process/supplier FMEA: N/A (beyond pure simulation boundary).

Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
Negative-electrode lithium plating surface	fast charge at 0.5C	absent	High (0.515 V (DFN) - (margin 5.6 mV))	Medium	Single-crystal cathode (0.8 μm, kinetic cut); charge termination
Thermal runaway from 4.7 V overcharge	self-heating at overcharge	0.5C to 4.7 V: T_max 298.22 K, triggered=false, dTdt max -8.6e-7 K/s	High	Low	Overcharge protection with hard cut at 4.7 V; do not
Excessive temperature at 45°C	charge vs cooling capacity	318.79 K vs 333.15 K limit (margin 14.4 K)	Medium	Low	Pouch double-face cold-plate cooling (h 25 W/m²K, A
Insufficient capacity / energy	electrode utilization, anode	10.65 Wh/kg vs 457.93 Wh/kg (margin +65.3)	Medium	Low	Anode 95 μm restores inventory (R4); single-crystal 1
Electrolyte oxidative decomposition	at high voltage	design overcharge thermal response (no exotherm)	Medium	Medium	4.7 V protection (no exotherm); 4.7 V threshold; direct stepping (saturation work
Particle fracture / cracking	of free cobalt (0.6 μm) and single-crystal cathode	Chen2020 — no simulation	Unverified	Medium (unverified)	Single-crystal cathode practice; prototype cycling vali

Conclusion

Highest-risk item: lithium plating margin (+5 mV DFN) — mitigation already implemented in design (anode kinetics + charge-window balance) with prototype validation recommended. Overcharge thermal runaway shows no trigger signal in the virtual model. Items without simulation signals are marked honestly as unverified rather than assumed safe.